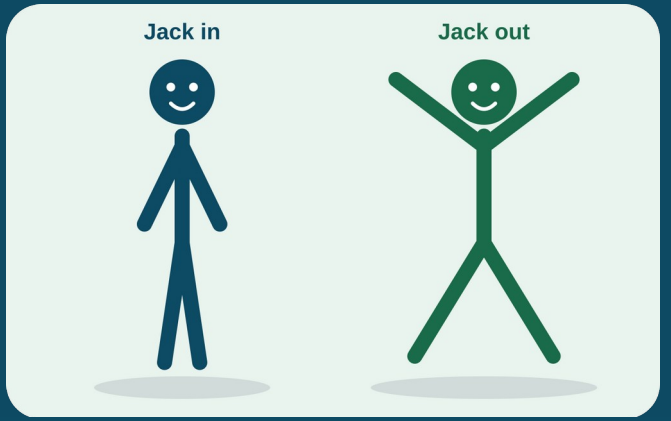
SAMPLE PRESENTATION

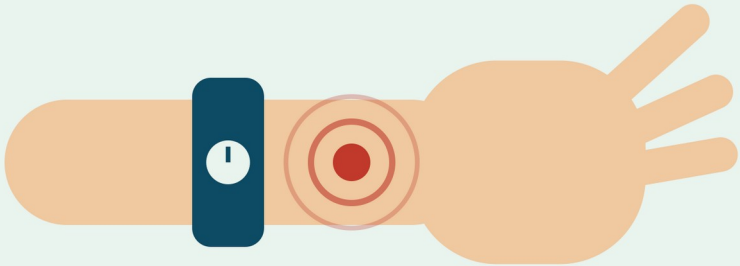
Does One Minute of Jumping Jacks Raise Heart Rate?

Sam & Riley · 7th Grade Science · Exercise Science Project



Why Heart Rate Matters

1. Heart rate is how many times your heart beats in one minute (bpm).
2. Scientists have estimated the highest heart rate a person can reach since the late 1930s **(Robergs & Landwehr, 2002)**.
3. Doctors and coaches use predicted maximum heart rate to plan exercise **(Tanaka et al., 2001)**.
4. We want to know how much one minute of jumping jacks raises a 12-year-old's heart rate.



Radial pulse: two fingers, wrist

Who benefits? Anyone who exercises, and students learning how their bodies respond.

WHY THIS WORKS Every fact from a source has a tag. Our own goal (point 4) has none.

Four Things to Know First



Resting heart rate

The beats per minute when you sit calmly.



Exercise raises it

Muscles need more oxygen, so the heart pumps faster.



Maximum heart rate

For healthy adults, about 208 − 0.7 × age (Tanaka et al., 2001). At age 12 that is roughly 200 bpm.



How to measure

Find the wrist pulse, count for 15 seconds, then multiply by 4.

WHY THIS WORKS Only the facts that help the audience understand the results. Only fact 3 needs a tag.

Related Research: Tanaka et al.

Tanaka, H., Monahan, K. D., & Seals, D. R. (2001). Age-predicted maximal heart rate revisited. *Journal of the American College of Cardiology*, *37*(1), 153–156. https://doi.org/10.1016/S0735-1097(00)01054-8

What they did

Combined data from 351 studies (18,712 people), then tested their equation on 514 healthy people in a lab.

What they found

Maximum heart rate is about 208 − 0.7 × age. Age predicts most of it, for men and women, active or not.

Link to our project

It tells us the rough ceiling for heart rate, so we can see how hard jumping jacks really are.

WHY THIS WORKS Full citation, a plain-language summary, and a link back to our experiment.

Related Research: Robergs & Landwehr

Robergs, R. A., & Landwehr, R. (2002). The surprising history of the “HRmax=220-age” equation. *Journal of Exercise Physiology Online*, *5*(2), 1–10.

What they did

Traced where the famous “220 – age” formula came from and checked how accurate such formulas are.

What they found

The formula was not built from original research, and age-based formulas can miss by about 7 to 11 beats per minute.

Link to our project

Predicted maximums are rough guides. That is why we measure real heart rates.

WHY THIS WORKS Second source on its own slide, with a connection to our experiment.

Question and Hypotheses



Research Question

How does one minute of jumping jacks affect a person's heart rate compared with resting?



Hypothesis

If a person does jumping jacks for one minute, then their heart rate will increase by at least 50%, because their muscles need more oxygen and the heart pumps faster to deliver it.



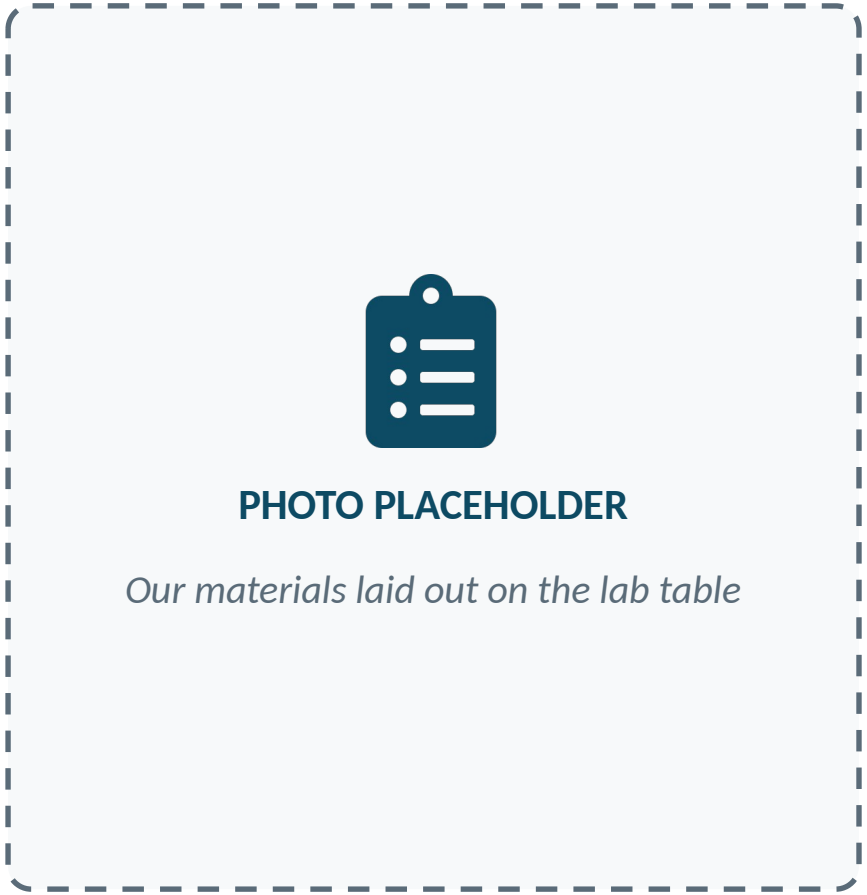
Null Hypothesis

There will be no significant difference between resting heart rate and heart rate after one minute of jumping jacks.

WHY THIS WORKS The question is testable. The hypothesis uses If / then / because. The null says "no difference."

Materials

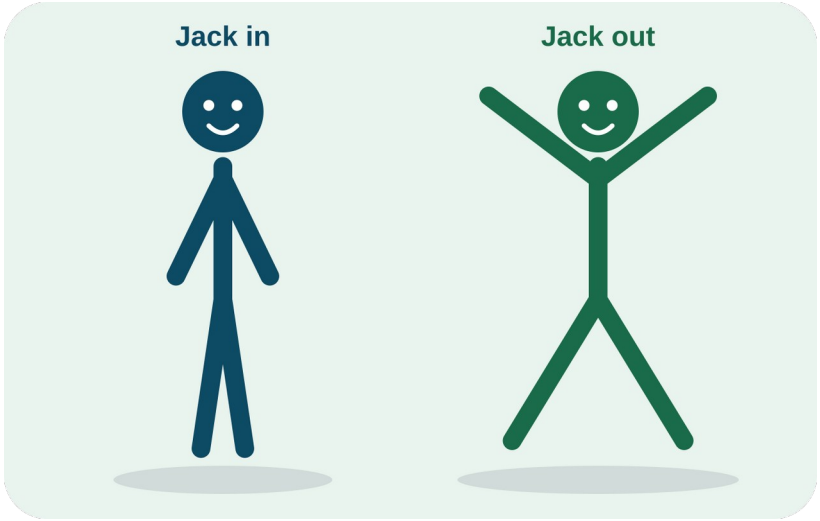
- Stopwatch (phone timer)
- Metronome app set to 60 beats per minute
- Data sheet on a clipboard
- Pencil
- Sneakers and open floor space



WHY THIS WORKS A complete bulleted list with specific items, and a photo of the real materials.

Procedure: 10 Trials

1. Riley sits quietly for 5 minutes.
2. Sam finds Riley's wrist pulse, counts for 15 seconds, and multiplies by 4. Record resting bpm.
3. Riley does jumping jacks for exactly 1 minute, 1 jack per second.
4. Riley sits. Within 5 seconds, Sam counts the pulse for 15 seconds and multiplies by 4. Record bpm.
5. Rest until the pulse is back to normal.
6. Repeat for 10 trials, always at the start of class.



Same pace every trial: 1 jack per second.

WHY THIS WORKS Numbered steps with exact times and counts. Another group could repeat this.

Variables and Controls

Independent Variable

Exercise condition: resting vs. 1 minute of jumping jacks

The one thing we change.

Dependent Variable

Heart rate in beats per minute (bpm)

What we measure: 15-second count × 4.

Controlled Variables

- Same person (Riley) every trial: so bodies match
- Same pace (metronome): so effort matches
- Same 5-minute seated rest: so every start is calm
- Same counting method: so measuring matches



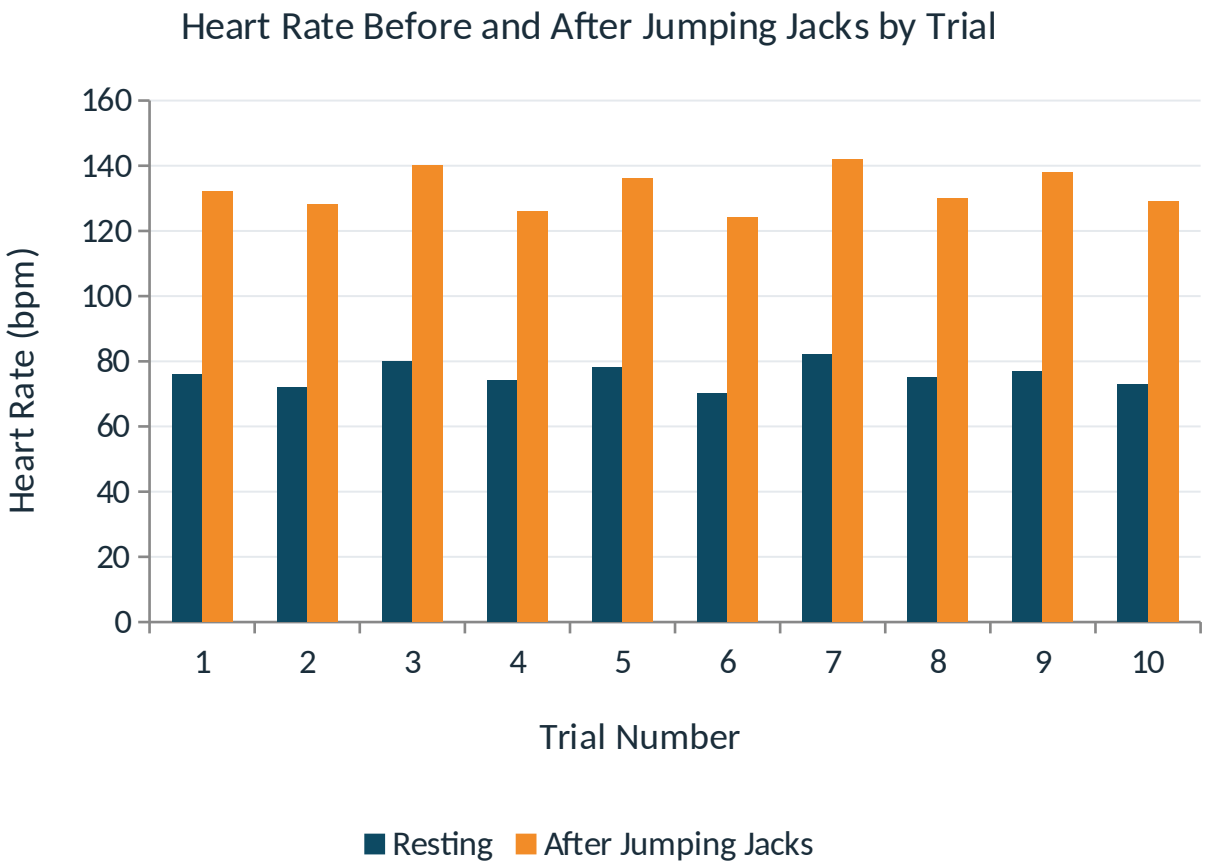
What we could not control: how much Riley slept and ate each day. We wrote it down so we can notice if it matters.

WHY THIS WORKS IV, DV, and four controls with reasons, plus an honest note about what we couldn't control.

Results: Table and Graph

SAMPLE DATA

Trial	Resting (bpm)	After Jumping Jacks (bpm)
1	76	132
2	72	128
3	80	140
4	74	126
5	78	136
6	70	124
7	82	142
8	75	130
9	77	138
10	73	129
Mean	**75.7**	**132.5**



WHY THIS WORKS Table and graph sit together, with Title Case labels, units, and a Mean row.

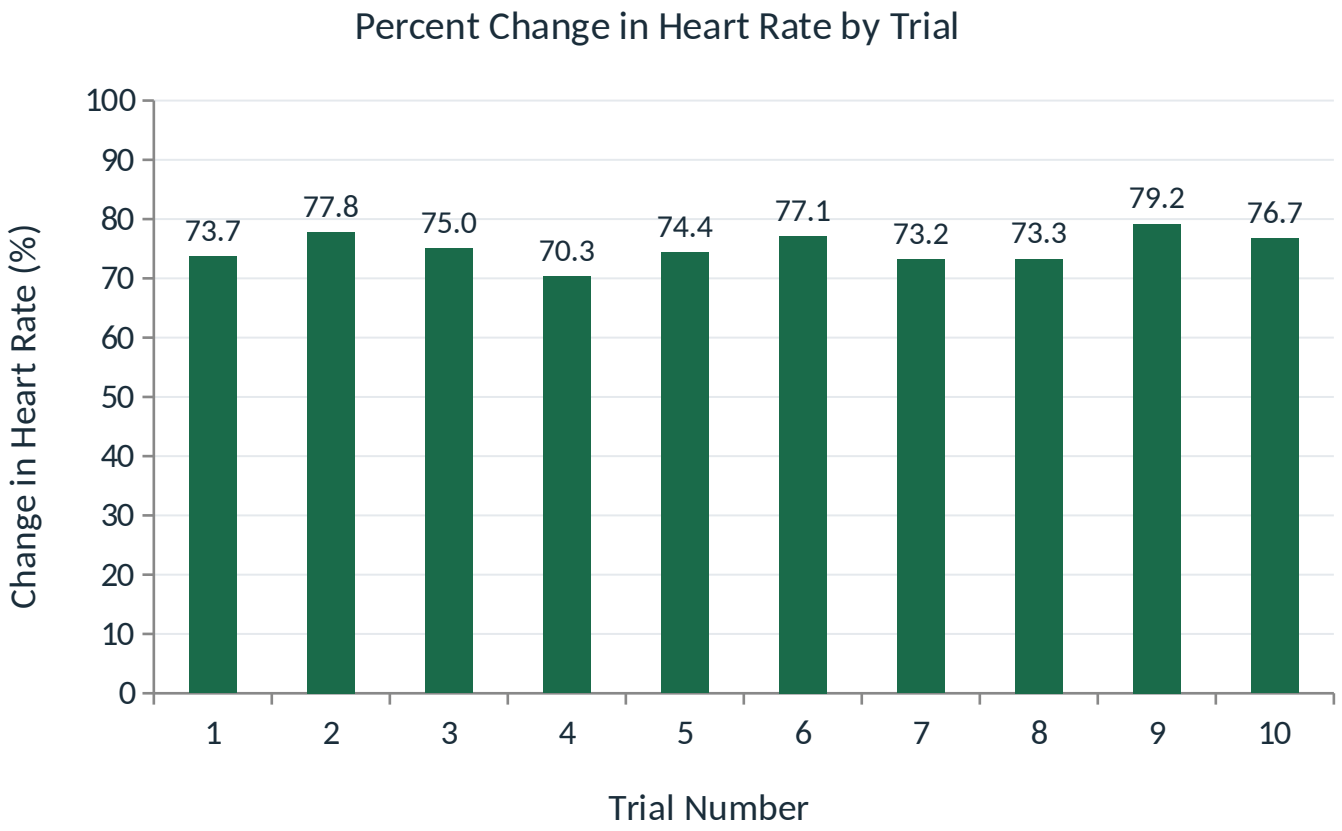
Percent Change and Statistics

SAMPLE DATA

+56.8 bpm
Mean increase

+75.0%
Mean percent change

p < 0.001
Paired t-test



Checked in Google Sheets with =T.TEST(range1, range2, 2, 1)

WHY THIS WORKS Percent change is fair across trials, and the statistics say whether the change is real.

Conclusion: Purpose, Claim, Evidence

Purpose

The purpose of this experiment was to investigate whether one minute of jumping jacks raises heart rate.

Claim

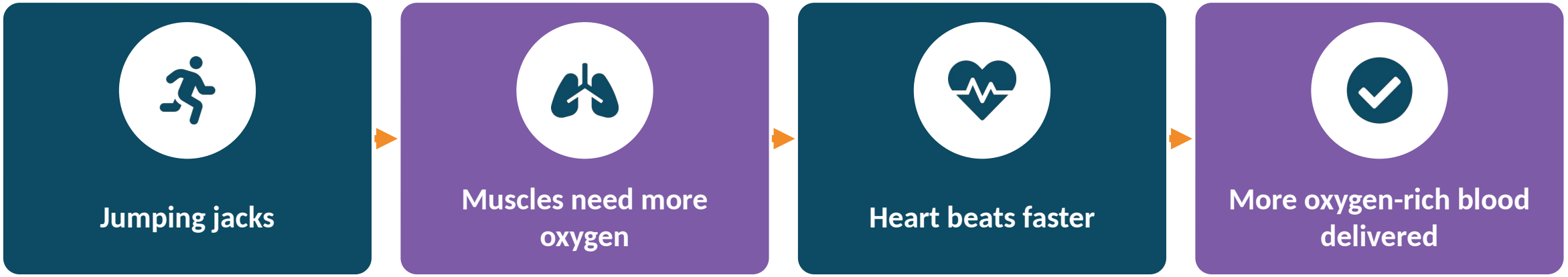
Based on the results, we found that heart rate was higher after jumping jacks. Our hypothesis was supported.

Evidence

According to the data, the average went from 75.7 bpm to 132.5 bpm, an increase of 56.8 bpm (75.0%). The paired t-test gave $p < 0.001$, so we reject the null hypothesis.

WHY THIS WORKS Each part starts with its sentence starter, and the evidence uses real numbers.

Conclusion: Reasoning



During exercise, working muscles need more oxygen. The heart beats faster to pump more oxygen-rich blood to them, which causes heart rate to rise. **Our mean after jumping jacks (132.5 bpm) is about two-thirds of the roughly 200 bpm maximum that the adult equation from Tanaka et al. (2001) predicts at age 12, so this was a moderate effort.**

WHY THIS WORKS Science vocabulary explains why it happened, and a source helps interpret the result.

Evaluation and Reflection

A

Sources of Error

We counted the pulse for 15 seconds and multiplied by 4, so one missed beat changes the result by 4 bpm. Counting started about 5 seconds after stopping, so the heart had begun to slow.

B

Improvements

Use a heart-rate monitor that records every beat. Have a third person call “go” and “stop” so counting starts right away.

C

Future Expansion

With millions of dollars and 5 years, we would track 1,000 students ages 10 to 14 with wearable sensors to compare fitness levels and recovery time.

WHY THIS WORKS No “human error.” Each problem is specific and comes with a fix.

References

Robergs, R. A., & Landwehr, R. (2002). The surprising history of the “HRmax=220-age” equation. *Journal of Exercise Physiology Online*, *5*(2), 1–10.

Tanaka, H., Monahan, K. D., & Seals, D. R. (2001). Age-predicted maximal heart rate revisited. *Journal of the American College of Cardiology*, *37*(1), 153–156. https://doi.org/10.1016/S0735-1097(00)01054-8



Alphabetical: R comes before T



Each entry matches a tag in the Introduction

WHY THIS WORKS APA format, alphabetical, with italics and DOI. Two real sources, as the rubric requires.

Questions?

Thank you! Sam & Riley

Sample presentation: data invented, sources real

